

Course Code-151609 Network Security**3 0 0 3****Unit-1.0: Introduction to Network Security**

Security fundamentals, Security threats and attack vectors, Vulnerability assessment, Security policies and risk management, Overview of cybersecurity frameworks.

Unit-2.0: Cryptography and Key Management

Symmetric key cryptography (DES, AES, RC4), Asymmetric key cryptography (RSA, ECC), Key exchange algorithms (Diffie-Hellman, Public Key Infrastructure - PKI), Hash functions (SHA, MD5), Digital signatures and certificates.

Unit-3.0: Authentication and Access Control

User authentication methods (passwords, multi-factor authentication, biometrics), Access control models (DAC, MAC, RBAC), Single Sign-On (SSO), OAuth, Kerberos authentication, Zero Trust security model.

Unit-4.0: Network Security Protocols

IPSec (AH, ESP), Secure Socket Layer (SSL) and Transport Layer Security (TLS), Virtual Private Networks (VPNs), Wireless security protocols (WEP, WPA, WPA2), Email security (PGP, S/MIME).

Unit-5.0: Firewalls, IDS, and Security Architectures

Types of firewalls (packet filtering, stateful, proxy-based), Intrusion Detection and Prevention Systems (IDS/IPS), Honeypots and deception technologies, Secure network design, Zero Trust Architecture.

Unit-6.0: Emerging Trends and Case Studies

Cloud security challenges and solutions, IoT security threats and countermeasures, AI and ML in cybersecurity, Blockchain for security, Case studies on recent cyber-attacks and defense strategies.

Text/Reference:

1. William Stallings, Network Security Essentials: Applications and Standards, Pearson.
2. Behrouz A. Forouzan, Cryptography and Network Security, McGraw Hill.
3. Charlie Kaufman, Radia Perlman, Mike Speciner, Network Security: Private Communication in a Public World, Pearson.
4. Kaufman, Perlman, and Speciner, Network Security: Private Communication in a Public World, Pearson.
5. Douglas R. Stinson, Cryptography: Theory and Practice, CRC Press.
6. Niels Ferguson & Bruce Schneier, Practical Cryptography, Wiley.
7. Joseph Migga Kizza, Guide to Computer Network Security, Springer.